Machine Learning

Non-parametric Algorithms: Parzen Window and k-NN Classifier

Agenda

- Parzen Window
- Defining Rn
- Two different approaches - fixed volume vs. fixed number of samples in a variable volume
- Example 3D hypercube
- The window function and estimation
- Critical parameters of the Parzen-window technique: window width and kernel
- Selecting Window
- Selecting Kernel

Parzen window

- The Parzen-window method (also known as Parzen-Rosenblatt window method) is a widely used non-parametric approach to estimate a probability density function p(x) for a specific point p(x) from a sample $p(x_n)$.

- It doesn't require any knowledge or assumption about the underlying distribution.

- A popular application of the Parzen-window technique is to estimate the class-conditional densities (or also often called 'likelihoods').

- Likelihoods, $p(x \mid \omega_i)$ in a supervised pattern classification problem from the training dataset (where p(x) refers to a multi-dimensional sample that belongs to particular class ω_i)).

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Where would this method be useful?

- Imagine that we are about to design a Bayes classifier for solving a statistical pattern classification task using Bayes' rule:

$$P(\omega_i \mid \boldsymbol{x}) = \frac{p(\boldsymbol{x} \mid w_i) \cdot P(\omega_i)}{p(\boldsymbol{x})}$$

$$\Rightarrow \textit{posterior probability} = \frac{\textit{likelihood} \cdot \textit{prior probability}}{\textit{evidence}}$$

- If the parameters of the the class-conditional densities (also called likelihoods) are known, it is pretty easy to design the classifier.
- Imagine we are about to design a classifier for a pattern classification task where the parameters of the underlying sample distribution are not known.
- Therefore, we wouldn't need the knowledge about the whole range of the distribution; it would be sufficient to know the probability of the particular point, which we want to classify, in order to make the decision.

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Parzen Window

- In parzen window, we are going to see how we can estimate this probability from the training sample.

- However, the only problem of this approach would be that we would seldom have exact values - if we consider the histogram of the frequencies for a arbitrary training dataset.

- Therefore, we define a certain region (i.e., the Parzen-window) around the particular value to make the estimate.

[1] *Parzen, Emanuel*. On Estimation of a Probability Density Function and Mode. The Annals of Mathematical Statistics 33 (1962), no. 3, 1065 1076.
[2] *Rosenblatt, Murray*. Remarks on Some Nonparametric Estimates of a Density Function. The Annals of Mathematical Statistics 27 (1956), no. 3, 832–837.

Defining the Region Rn

- The basis of this approach is to count how many samples fall within a specified region Rn (or "window"). Our intuition tells us, that (based on the observation), the probability that one sample falls into this region is:

$$p(x) = \frac{\#\ of\ samples\ in\ R}{total\ samples}$$

- To tackle this problem from a more mathematical standpoint to estimate "the probability of observing k points out of n in a Region R ", we consider **a binomial distribution:**

$$p_k = \begin{bmatrix} n \\ k \end{bmatrix} \cdot p^k \cdot (1-p)^{n-k}$$

- Make the assumption that in a binomial distribution, the probability peaks sharply at the mean

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$$E[k] = n \cdot p \ \sim \ k = n \cdot p$$

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Defining the Region Rn

- And if we think of the probability as a continuous variable, we know that it is defined as:

$$p(\mathbf{x}) = \int_R dx = p(\mathbf{x}) \cdot V,$$

where V is the volume of the region R, and if we rearrange those terms, so that we arrive at the following equation:

$$\frac{k}{n} = p(\mathbf{x}) \cdot V$$

$$\Rightarrow p(\mathbf{x}) = \frac{k/n}{V}$$

- This equation above (i.e, the “probability estimate”) lets us calculate the probability density of a point x by counting how many points k fall in a defined region (or volume).

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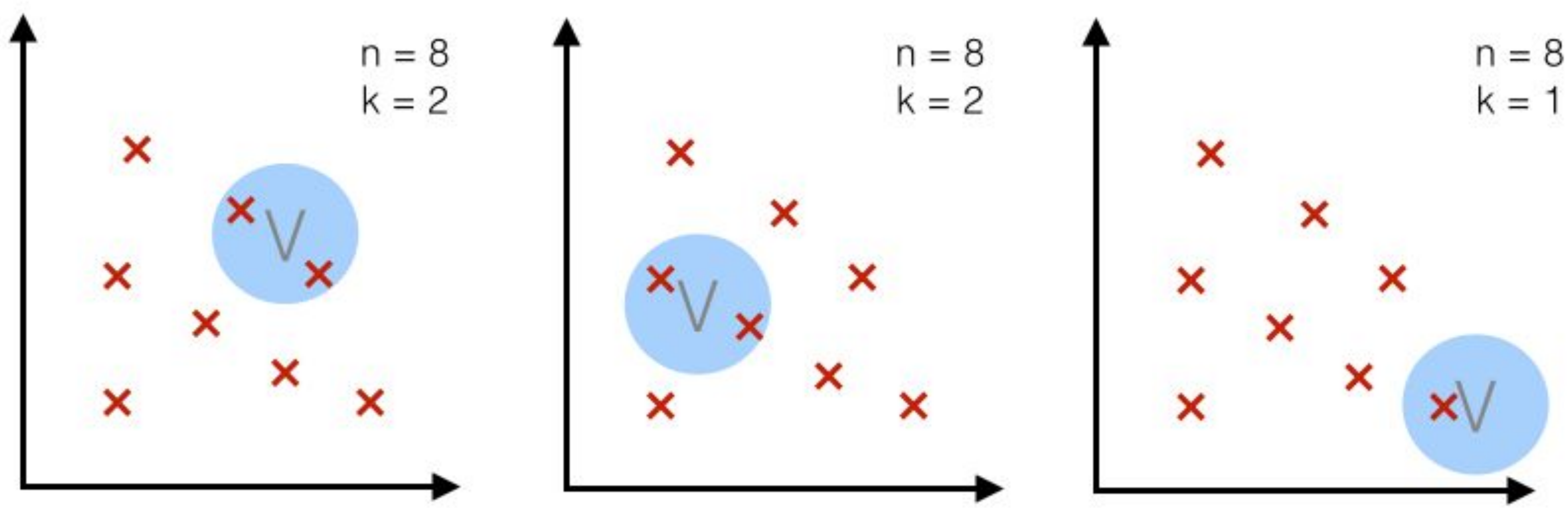
Two different approaches - fixed volume vs. fixed number of samples in a variable volume

Case 1 - fixed volume:

- For a particular number n (= number of total points), we use volume V of a fixed size and observe how many points k fall into the region.



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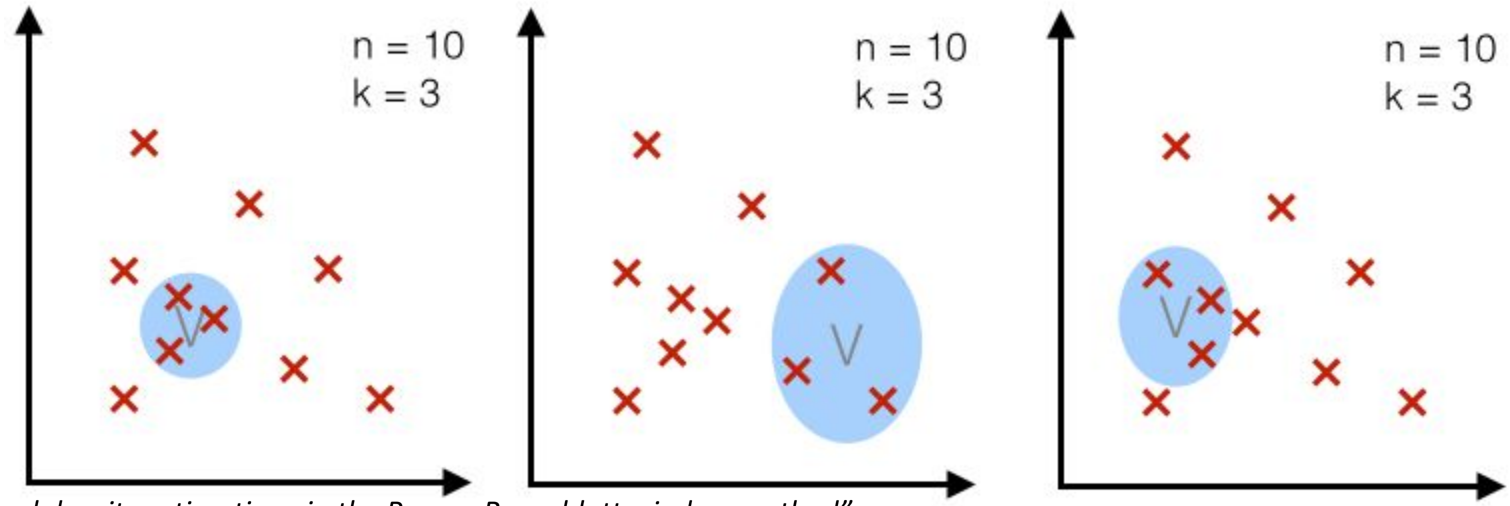
Two different approaches - fixed volume vs. fixed number of samples in a variable volume

Case 2 - fixed *k*:

- For a particular number n (= number of total points), we use a fixed number k (number of points that fall inside the region or volume) and adjust the volume accordingly..



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Example 3D-hypercubes

- To illustrate this with an example and a set of equations, let us assume this region R_n is a hypercube.

- The volume of this hypercube is defined by $V_n = h_n^d$, where h_n is the length of the hypercube, and d is the number of dimensions.

- For an 2D-hypercube with length 1, for example, this would be $V_1 = 1^2$ and for a 3D hypercube $V_1 = 1^3$, respectively.

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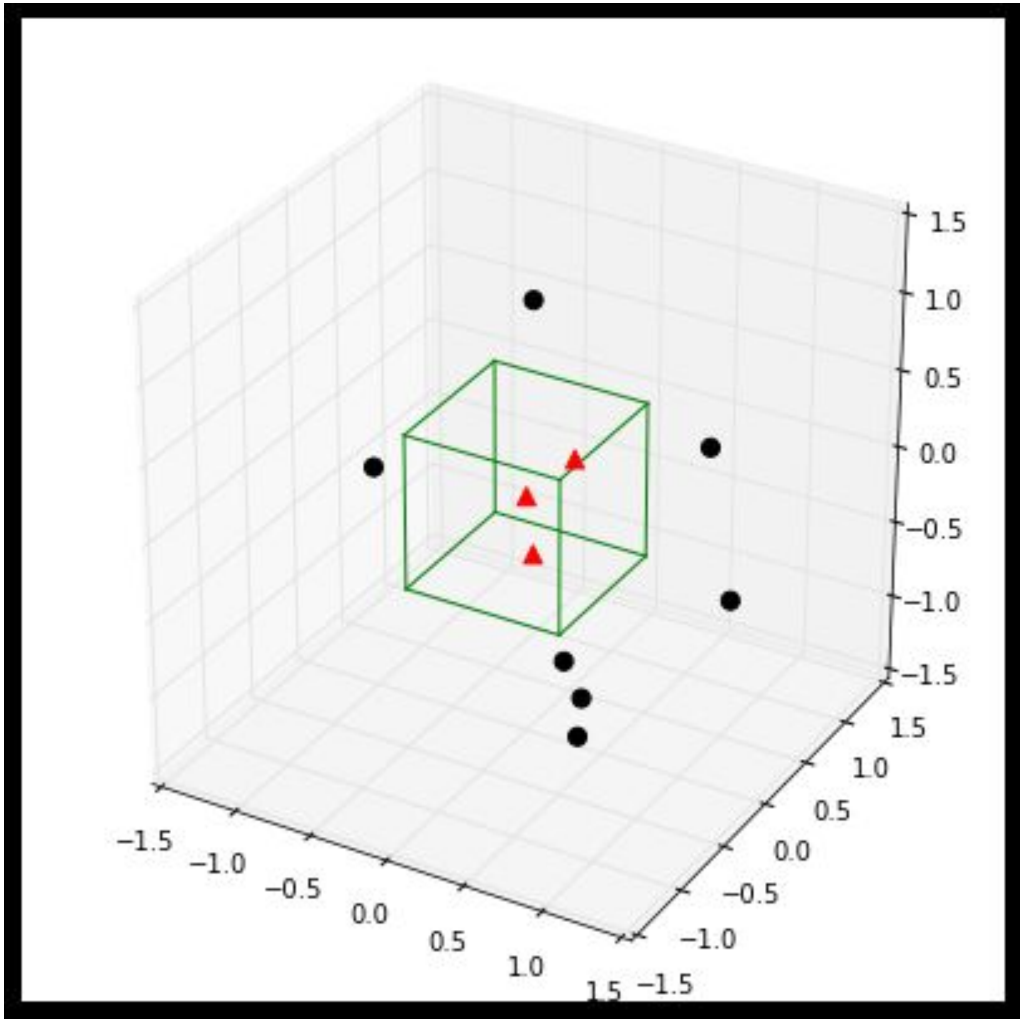
Example: A typical 3-dimensional unit hypercube ($h_1 = 1$) representing the region R_1, and 10 sample points, where 3 of them lie within the hypercube (red triangles), and the other 7 outside (black dots).



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The window function

- Once we visualized the region R1 like above, it is easy and intuitive to count how many samples fall within this region, and how many lie outside.
- To approach this problem more mathematically, we would use the following equation to count the samples k_n within this hypercube, where φ is our so-called window function.

$$\phi(\boldsymbol{u}) = \begin{cases} 1 & |u_j| \leq 1/2 \,; \quad j = 1, \ldots, d \\ 0 & \textit{otherwise} \end{cases}$$

for a hypercube of unit length 1 centred at the coordinate system's origin.

- If we extend on this concept, we can define a more general equation that applies to hypercubes of any length h_n that are centered at *x*:

$$k_n = \sum_{i=1}^{n} \phi\left(\frac{\mathbf{x}-\mathbf{x}_i}{h_n} \right)$$

$$\text{where } \boldsymbol{u} = \left(\frac{\mathbf{x}-\mathbf{x}_i}{h_n} \right)$$

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Parzen-window estimation

- we can now formulate the Parzen-window estimation with a hypercube kernel as follows:

$$p_n(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{h^d} \phi \left[\frac{\mathbf{x}-\mathbf{x}_i}{h_n} \right]$$

where

$$h^d = V_n \quad \text{and} \quad \phi \left[\frac{\mathbf{x}-\mathbf{x}_i}{h_n} \right] = k$$

- And applying this to our unit-hypercube example above, for which 3 out of 10 samples fall inside the hypercube (into region *R*), we can calculate the probability *p(**x**)* that ***x*** samples fall within region *R* as follows:

$$\mathbf{x} = \frac{k/n}{h^d} = \frac{3/10}{1^3} = \frac{3}{10} = 0.3$$

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Critical parameters of the Parzen-window technique: window width and kernel

The two critical parameters in the Parzen-window techniques are

1) the window width: Which window size should we choose (i.e., what should be the side length h of our hypercube)?

2) the kernel: Most commonly, either a hypercube or a Gaussian kernel is used for the window function. But how do we know which is better?

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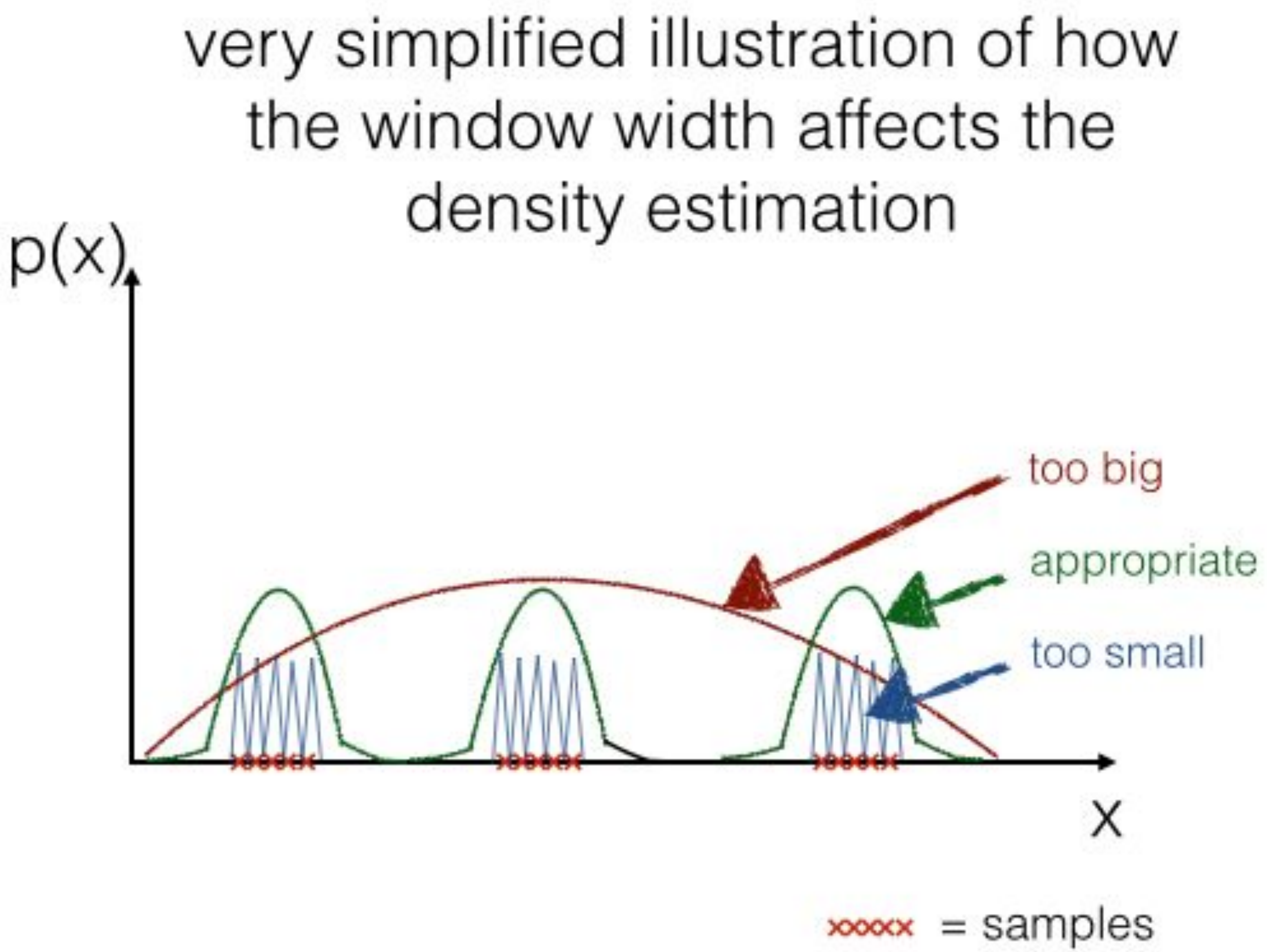
Selecting the window width



If we would choose a window width that is “too small”, this would result in local spikes, and a window width that is “too big” would average over the whole distribution.

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Selecting the kernel

- Hypercube or a Gaussian kernel?
  - It really depends on the training sample.

- Intuitively, it would make sense to use a Gaussian kernel for a data set that follows a Gaussian distribution.

  - *But remember, the whole purpose of the Parzen-window estimation is to estimate densities of a* ***unknown distribution!***

- ***Gaussian kernel instead of the hypercube:***
  - *simply swap the terms of the window function, which we defined above for the hypercube.*

Gaussian Kernel

- **The Parzen-window Gaussian kernel:**

$$\frac{1}{(\sqrt{2\pi})^d h_n^d} exp\left[-\frac{1}{2}\left(\frac{\mathbf{x}-\mathbf{x}_i}{h_n}\right)^2\right]$$

- **The Parzen-window estimation would then look like this:**

$$p_n(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{h^d}\phi\left[\frac{1}{(\sqrt{2\pi})^d h_n^d} exp\left[-\frac{1}{2}\left(\frac{\mathbf{x}-\mathbf{x}_i}{h_n}\right)^2\right]\right]$$

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Thank You:
Question?